REAL AI PROMPT LOG

Rossmann Christmas Footfall Prediction

Based only on prompts actually typed in this ChatGPT conversation

Authenticity note
The prompts below reproduce the user's actual wording from this conversation, including informal wording and spelling. Annotations explain the analytical decision that followed. No feature-selection, model-selection or hyperparameter-tuning prompt has been invented. Where the group later made modelling decisions outside this chat, those prompts must be added by the member who actually used them.

1. Business problem and problem framing

Prompt 1

Exact prompt typed in ChatGPT
This a business analytics assignment. We need to identify a business problem and work on it. I want to work on any problem related to footfall increase during Diwali. What can I work on

Purpose: Identify a machine-learning business problem connected with festive retail footfall.

AI recommendation / outcome: The discussion converted the broad objective of increasing footfall into a measurable ML task: predicting footfall or classifying high/low-footfall days using promotions, calendar variables and store characteristics.

Decision made: Use a high/normal footfall classification problem rather than treating 'increase footfall' as the target itself.

Reasoning / annotation: 'Increase footfall' is a business objective, not a directly observed target variable. A classification outcome is measurable and can later support staffing, promotion and inventory decisions.

Prompt 2

Exact prompt typed in ChatGPT
What can be the business problem statement here

Purpose: Convert the broad topic into a formal business problem statement.

AI recommendation / outcome: The proposed statement focused on variation in festive footfall and the resulting difficulty in staffing, inventory, promotion and resource planning.

Decision made: Adopt a predictive festive-footfall business problem.

Reasoning / annotation: The framing connects the ML output to a clear managerial use case rather than presenting modelling as an isolated technical exercise.

Prompt 3

Exact prompt typed in ChatGPT
okay do it .

Purpose: Confirm the choice of the classification-style festive-footfall project.

AI recommendation / outcome: The project was formalised around predicting high vs low customer footfall using ML.

Decision made: Proceed with a classification formulation.

Reasoning / annotation: Classification fits the assignment's requirement to compare models using Accuracy, Precision, Recall, F1 and ROC-AUC.

Prompt 4

Exact prompt typed in ChatGPT
yes please

Purpose: Find a dataset capable of supporting the proposed footfall problem.

AI recommendation / outcome: Rossmann Store Sales was recommended because it contains daily Customers, Promo, holiday and store-level fields.

Decision made: Select Rossmann Store Sales.

Reasoning / annotation: The dataset includes direct customer counts, which are more appropriate for a footfall project than using sales as a proxy.

Prompt 5

Exact prompt typed in ChatGPT
if we are going with the euopean data set i think we can take christmas ? otherwise it make less sense for me to go for europen data set with planning for indian festival

Purpose: Check whether the festival context should match the geography of the dataset.

AI recommendation / outcome: The project was reframed from Diwali to Christmas.

Decision made: Use Christmas rather than Diwali.

Reasoning / annotation: A European retail dataset paired with a European Christmas-shopping context is more internally consistent and easier to justify academically.

Prompt 6

Exact prompt typed in ChatGPT
okay check it

Purpose: Verify whether Rossmann has enough Christmas-period observations and suitable fields.

AI recommendation / outcome: The dataset was checked for date coverage and Christmas relevance; the analysis was focused on the pre-Christmas shopping period rather than Christmas Day.

Decision made: Use 1-24 December and open stores as the core Christmas shopping window.

Reasoning / annotation: Christmas Day and other closure days can produce zero customers because the store is closed, not because demand is low. The pre-Christmas period better represents active shopping demand.

Prompt 7

Exact prompt typed in ChatGPT
okay go ahead

Purpose: Proceed from topic selection to the complete project framework.

AI recommendation / outcome: A framework was prepared covering business problem, objectives, metadata, target, EDA, preprocessing and planned ML models.

Decision made: Use the Rossmann Christmas footfall classification framework.

Reasoning / annotation: This provides the structure required by the assignment before coding begins.

2. Guideline-check and Python implementation prompts

Prompt 8

Exact prompt typed in ChatGPT
did you read guidelines ?

Purpose: Verify that the work follows the actual assignment instructions.

AI recommendation / outcome: The assignment requirements were re-checked: problem definition, EDA, preprocessing, GenAI comparison, feature selection, model building, metrics, model versions, actionable solution and prompt logs.

Decision made: Reorganise the project explicitly around the assignment steps.

Reasoning / annotation: This avoids producing technically correct analysis that misses assessment-specific evidence such as prompt logs, model iterations and AI-use justification.

Prompt 9

Exact prompt typed in ChatGPT
yes

Purpose: Confirm moving ahead using the guideline-based structure.

AI recommendation / outcome: The project was restructured into stages aligned to the rubric.

Decision made: Proceed according to the formal rubric sequence.

Reasoning / annotation: This improves traceability and makes each deliverable easier to map to the grading requirements.

Prompt 10

Exact prompt typed in ChatGPT
okay move ahead'

Purpose: Proceed with Step 2: descriptive analytics, EDA and preprocessing.

AI recommendation / outcome: Actual Rossmann data was analysed for missingness, Christmas-period summaries, promotions, day-of-week and store-type patterns.

Decision made: Move from planning into dataset-driven EDA and preprocessing.

Reasoning / annotation: The assignment requires executed analytical work rather than only conceptual discussion.

Prompt 11

Exact prompt typed in ChatGPT
but pre processing has to be done on phython code i think we are learning ML on phyton code , just check the guidelines

Purpose: Check whether preprocessing must be executed in Python rather than only described.

AI recommendation / outcome: The guideline explicitly requires Python/Jupyter/Colab code for descriptive analytics, preprocessing, feature selection, model building and evaluation.

Decision made: Implement preprocessing in Python.

Reasoning / annotation: This is necessary for the required executed notebook and HTML submission.

Prompt 12

Exact prompt typed in ChatGPT
okay do it

Purpose: Create the actual Python preprocessing workflow.

AI recommendation / outcome: An executable Jupyter notebook was created with loading, merging, filtering, feature engineering, target creation, chronological splitting, imputation, encoding and scaling.

Decision made: Use an executable notebook rather than report-only preprocessing.

Reasoning / annotation: This makes the workflow reproducible, visible and compliant with the submission requirements.

3. Real Step 2 cross-GenAI prompt

Prompt 13

Exact prompt typed in ChatGPT
I am analysing the Rossmann Store Sales dataset to predict high customer footfall during the Christmas shopping period. The dataset contains Customers, Promo, Date, DayOfWeek, Open, StateHoliday, SchoolHoliday, StoreType, Assortment, CompetitionDistance, competition opening dates, Promo2 and PromoInterval. Recommend appropriate EDA, missing-value treatment, feature engineering, categorical encoding, leakage prevention, class-imbalance handling, and train-test strategy for a classification model. Explain the reasoning for every recommendation.

Purpose: Obtain comparable EDA and preprocessing recommendations from GenAI tools as required by the assignment.

AI recommendation / outcome: ChatGPT recommended Christmas-specific EDA, structural treatment of Promo2 missingness, median imputation for sparse numeric missingness, DaysToChristmas and competition features, one-hot encoding, leakage prevention, checking imbalance before SMOTE, and chronological validation.

Decision made: Use the recommendations only after checking them against the real Rossmann data.

Reasoning / annotation: The assignment requires comparison of 2-3 GenAI tools, but analytical choices should be justified with dataset evidence rather than copied automatically.

4. Google Gemini response supplied by the user

Exact common prompt sent to Gemini
I am analysing the Rossmann Store Sales dataset to predict high customer footfall during the Christmas shopping period. The dataset contains Customers, Promo, Date, DayOfWeek, Open, StateHoliday, SchoolHoliday, StoreType, Assortment, CompetitionDistance, competition opening dates, Promo2 and

PromoInterval. Recommend appropriate EDA, missing-value treatment, feature engineering, categorical encoding, leakage prevention, class-imbalance handling, and train-test strategy for a classification model. Explain the reasoning for every recommendation.

Gemini response supplied in this chat

To predict high customer footfall during the Christmas shopping period (a binary classification task: High Footfall vs. Normal/Low Footfall), here is a structured machine learning framework designed for the Rossmann Store Sales dataset.

1. Exploratory Data Analysis (EDA)

Temporal Footfall Trends: Plot daily and weekly customer counts over time, specifically highlighting November-December, to analyze the magnitude and duration of the Christmas traffic spike.

Promo & Holiday Impact Analysis: Compare distributions of Customers across Promo, StateHoliday (specifically a = public holiday, b = Easter, c = Christmas), and SchoolHoliday using box plots or violin plots.

Store Type & Assortment Interaction: Group stores by StoreType and Assortment to evaluate which store formats experience the largest surge in holiday traffic.

Competition Analysis: Scatter plot Customers against CompetitionDistance and compare sales/footfall before vs. after competition open dates.

Promo2 Long-term Effect: Analyze whether long-term continuous promotions (Promo2, PromoInterval) positively or negatively affect peak holiday traffic.

Reasoning: EDA reveals temporal seasonality, store-level baselines, and promotional responsiveness, helping inform feature design and data split strategies.

2. Missing-Value Treatment

CompetitionDistance: Impute missing values with a high constant (e.g., 999999 or maximum distance +1) or the median distance per StoreType.

Reasoning: Missing distance generally indicates that there is no nearby competitor; assigning a large distance logically represents an isolated store environment.

CompetitionOpenSinceMonth & CompetitionOpenSinceYear: Impute with a distinct placeholder (e.g., 0) and create a binary indicator flag CompetitionOpen_Missing.

Reasoning: Missing values mean no competition open date is recorded. A flag allows tree-based models to treat missing competition data as a specific signal.

Promo2SinceWeek, Promo2SinceYear, PromoInterval: Fill missing values with 0 (or 'None' for PromoInterval).

Reasoning: Missing entries correspond directly to Promo2 == 0 (stores not participating in Promo2).

3. Feature Engineering
Target Labeling (High Footfall): Define a binary target Is_High_Footfall. Compute a store-specific quantile threshold (e.g., 75th or 80th percentile of daily Customers when Open == 1) to mark 1 for high footfall days and 0 otherwise.

Reasoning: Store sizes and customer baselines vary drastically. A global threshold would misclassify small, busy stores as low footfall; store-relative thresholds measure true peak traffic.

Calendar & Holiday Features:
Extract Month, Day, WeekOfYear, DayOfWeek, Is_Weekend, and Days_To_Christmas (days remaining until Dec 25).
Create lead/lag flags for holidays: Is_Pre_Holiday (1-3 days before StateHoliday or SchoolHoliday) and Is_Post_Holiday.

Reasoning: Christmas shopping ramps up significantly in the days leading up to holidays.

Competition Age: Calculate CompetitionAgeInMonths = (Date - CompetitionOpenDate).

Reasoning: Newer competitors impact footfall differently than long-established ones.

Promo2 Activity Flag: Derive Is_Promo2_Active by checking if the current date falls within PromoInterval and is past Promo2SinceWeek/Year.

Reasoning: Raw PromoInterval strings cannot be directly ingested by models; a dynamic boolean flag correctly captures active promotional periods.

Historical Aggregations (Lags & Rolling Means): Compute historical average customers per store by DayOfWeek, Month, and Promo status using expanding or rolling windows (excluding future dates).

Reasoning: Historical traffic patterns per store are strong predictors of future traffic demand.

4. Categorical Encoding
StoreType & Assortment: Target Encoding (smoothed out-of-fold target encoding) or One-Hot Encoding (since cardinality is low: 4 store types, 3 assortment levels).

Reasoning: Low-cardinality categories are cleanly represented via One-Hot Encoding without increasing dimensionality excessively.

StateHoliday: Map 0/'0' to 0, a to 1, b to 2, c to 3 (Ordinal Encoding) or One-Hot Encoding.

Reasoning: Captures distinct holiday types explicitly.

Store ID: Frequency Encoding or Target Encoding based on historical mean footfall. Alternatively, rely on store-level aggregated features rather than raw Store ID if generalizing to new stores.

Reasoning: Directly passing raw IDs can cause overfitting or poor generalization in tree models.

5. Leakage Prevention
Target Leakage Guardrails:
Do NOT use Sales as a predictor feature for predicting Customers/Footfall.

Reasoning: Sales and Customers occur simultaneously in real time. Using sales to predict footfall is a direct data leakage violation.

Rolling & Lag Window Boundaries: Ensure all rolling features (e.g., 7-day average customers) use strict lag shifts (shift(1) or greater) so they only aggregate past data strictly prior to the prediction date.

Out-of-Fold Aggregations: Calculate any target-encoded features strictly inside training folds during cross-validation to prevent validation target signals from bleeding into training features.

6. Class-Imbalance Handling
Threshold Adjustment: Tune the classification decision threshold (e.g., optimizing Precision-Recall AUC or F-score prioritizing Recall) rather than using the default 0.5 cutoff.

Reasoning: Predicting high footfall often requires higher Recall to ensure stores are adequately staffed and stocked during surge days.

Cost-Sensitive Learning: Set scale_pos_weight in LightGBM/XGBoost or use class_weight='balanced' in Random Forest/Logistic Regression.

Reasoning: Automatically penalizes false negatives higher than false positives, aligning the loss function with operational goals.

Sampling Techniques (Optional): Apply SMOTE or random undersampling only within training folds if severe class skew exists.

7. Train-Test Strategy
Temporal Train-Validation-Test Split (Out-of-Time Split):
Train Set: All historic data up to late November of year N-1 and preceding years.
Validation Set: December of year N-1 (Christmas shopping period of the previous year).
Test Set: Christmas shopping period of the final year.
Time-Series Group Cross-Validation: Use TimeSeriesSplit or expanding-window time-based CV folds.

Reasoning: Random K-Fold cross-validation leaks future seasonal trends into past evaluations and creates look-ahead bias. Evaluating on a historical Christmas period mirrors the exact real-world deployment scenario.

5. Real ChatGPT-Gemini comparison and decisions

Decision area	ChatGPT	Gemini	Final decision	Reasoning
Target	ChatGPT: store-specific	Gemini: P75/P80	P75 selected	Provides a relative high-footfall definition while

	P75			retaining enough positive cases.
Promo2 missingness	Structural missingness	Fill 0/None when Promo2=0	Structural handling + Promo2Active	Missingness carries business meaning; it should not be treated as accidental.
CompetitionDistance	Median imputation	Large constant or StoreType median	Median imputation	Only a very small fraction is missing; extreme artificial values could distort the feature.
Weekend	Retain DayOfWeek	Create Is_Weekend	DayOfWeek retained	Sunday openings are unusual and should not be collapsed with Saturday.
Sales leakage	Exclude same-day Sales	Exclude same-day Sales	Excluded	Sales is not available before the day's footfall outcome.
SMOTE	Only if imbalance is severe	Only if needed	Not used initially	The constructed target was not severely imbalanced.
Validation	Chronological split	Out-of-time/time-series split	2013 development, 2014 holdout	Maintains future direction and avoids look-ahead bias.

6. Real notebook/EDA support prompts

Prompt 14

Exact prompt typed in ChatGPT
how to do in collab ?

Purpose: Run the notebook in Google Colab.

AI recommendation / outcome: Upload the notebook and CSV files and change the paths to /content/.

Decision made: Use Colab for execution.

Reasoning / annotation: The assignment requires an executed notebook; Colab provides an accessible environment.

Prompt 15

Exact prompt typed in ChatGPT
eda is done before pre processing or after ?

Purpose: Clarify the correct analytical order.

AI recommendation / outcome: EDA should follow minimal preparation but precede final ML preprocessing.

Decision made: Place EDA after merge/date filtering and before imputation/encoding/scaling.

Reasoning / annotation: EDA should inform preprocessing decisions rather than be performed only after all transformations.

Prompt 16

Exact prompt typed in ChatGPT
basic calender feature enginnering above this i have to paste EDA codes

Purpose: Locate the EDA section in the notebook.

AI recommendation / outcome: EDA should not be inserted before the basic feature preparation required to define the Christmas sample.

Decision made: Keep calendar features first.

Reasoning / annotation: The EDA charts depend on the Christmas-window variables.

Prompt 17

Exact prompt typed in ChatGPT
befoe competitoon related features?

Purpose: Check the exact insertion point.

AI recommendation / outcome: The EDA section should be placed after the Christmas modelling window creates `christmas_open`, not before competition/Promo2 preparation in the existing notebook structure.

Decision made: Insert EDA after Step 8 in the supplied notebook.

Reasoning / annotation: That is the first point where the EDA dataframe exists and the supplied chart code can run safely.

Prompt 18

Exact prompt typed in ChatGPT
so after the output i have to add ?

Purpose: Confirm exact Colab placement.

AI recommendation / outcome: Yes, after the Step 8 output and before target construction.

Decision made: Insert a text heading and EDA code cells there.

Reasoning / annotation: This keeps the notebook readable and preserves the intended analysis order.

7. Important gap: model-selection and tuning prompts

Do not fabricate these

This chat contains real prompts for problem framing, dataset choice, EDA, preprocessing, Python implementation and GenAI comparison. It does NOT contain a genuine teammate prompt that asked an AI to perform the final V1/V2 feature-selection comparison, select Random Forest V1, or tune its hyperparameters. The final project report contains those decisions, but a decision recorded in the report is not the same thing as an AI prompt. If another member used ChatGPT/Gemini for feature selection, model selection or tuning, paste their exact prompt and response into this PDF before final submission.

8. Template for the other members to add their real prompts

Use this format for every genuine Step 3/4 prompt

Exact prompt:
[Paste exactly what the member typed]

AI tool:
[ChatGPT / Gemini / other]

AI recommendation:
[What it recommended]

Change implemented:
[What code/model/feature/hyperparameter was changed]

Metric before:
[Validation result before change]

Metric after:
[Validation result after change]

Decision:
[Accepted / rejected]

Reasoning:
[Why the result justified the decision]